MILITARY SPECIFICATION SHEET

FILTER ELEMENT, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 3000 PSI, 5 MICRON ABSOLUTE, -65° TO +275°F, F14 AIRCRAFT, 30 GPM

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-8815.

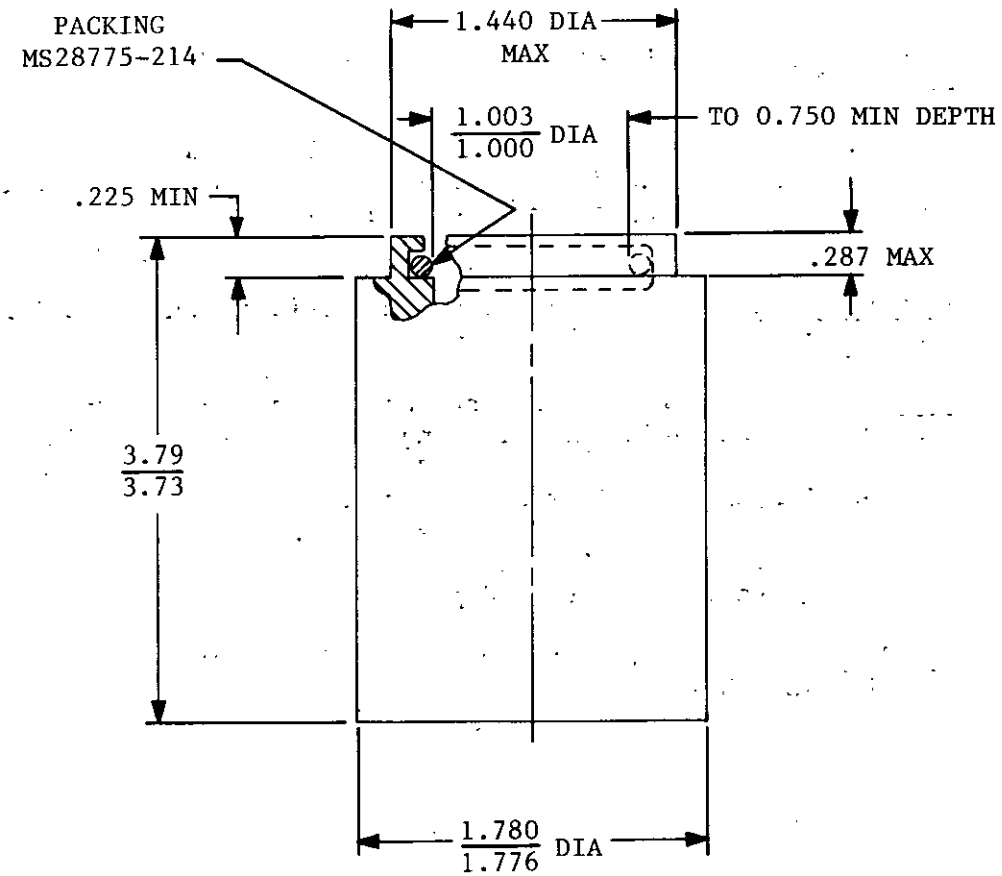


Figure 1. Dimensions

FSC 1650

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NOTES:

1. Dimensions in inches unless otherwise specified.

2. Tolerances, decimals ± .010, angles ± 1°.

3. Finish all over 100 except as noted. Surface texture in accordance with ASA B46.1 - 1962 (ANSI).

4. Material shall be in accordance with the procurement specification.

REQUIREMENTS:

Envelope: See Figure 1.

Pressure Drop: Maximum pressure drop at rated flow across the filter element shall not exceed 15 psid at 100 ± 10°F.

Contaminants: See Table I.

Degree of Filtration: The filter element shall remove 100 percent of all contaminant larger than 5 microns and 95 percent of all test contaminant. The filter efficiency shall be a minimum of 95 percent.

Element Collapse: The filter element shall withstand a differential pressure of 4500 psid.

Cold Start: The element shall withstand 3000 psid differential pressure across the element for the flow cycles and fluid temperature as specified in the general specification. The element pressure drop shall not exceed 15 psid following the cold start test at 100 ± 10°F.

Maximum Particle Passed: The maximum particle passed shall be no greater than 5 microns.

Flow Fatigue: The filter element shall withstand the following flow fatigue pressure cycles at a cycling rate of 100 cpm maximum.

90,000 cycles at 20 psid pressure drop
8,000 cycles at 50 psid pressure drop
2,000 cycles at 100 psid pressure drop
500 cycles at 3000 psid pressure drop

Part Number: The part number shall consist of the basic number of this specification sheet.

Example:

M8815/23(AS)

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TABLE I. CONTAMINANTS

Flow Rate (GPM)	Degree of Filtration Add Rate		Maximum Particle Add Rate		Pressure Buildup (at 100 ± 10°F)		Migration Analysis Value	
							Maximum Allowable (gms)	
	F-9 Beads (gms)	AC Dust (gms)	F-9 Beads (gms)	Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms)	After Cold Start 4.7.2.3	After Vibration 4.7.2.8.3.2
30.0	0.141	0.141	0.042	0.042	0.250	3.3	0.0350	0.0070

Maximum Weight: The maximum weight of the filter element shall be 0.50 pounds.

Intended Use: The filter element covered by this specification sheet is intended for use as specified in paragraph 6.1 of the general specification.

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TESTS:

The following tests shall be conducted on filter element number 1, in addition to the tests specified in the general specification. (See Table I of the general specification).

MIL-H-83282 Immersion: Following the immersion test specified in the general specification, the filter element shall be immersed in hydraulic fluid conforming to MIL-H-83282 for a period of 72 hours at a fluid temperature of 275°F and 72 hours at a fluid temperature of -65oF. All internal parts shall be in contact with the test fluid during this period. After immersion, the filter shall remain in this fluid at room temperature until further tests are conducted.

Element Bonding Test: Following the flow fatigue test specified in Table I of the general specification the bonding test shall be conducted to verify the element assembly integrity and whether degradation has occurred during prior testing. Support the element end cap (closed end) by means of a rigidly mounted fixture. The fixture shall distribute the load around the element end cap periphery. Apply an axial tensile load to the header (open end) for 15 seconds and then gradually reduce the load to zero. The load shall be 25 $^{+5}_{-0}$ pounds.

Repeat the load cycle five times. After five cycles inspect the element. There shall be no element deformation or bonding failures.

Preparing Activity

Navy - AS

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-F-8815/23(AS) Filter Element, Noncleanable, Fluid Pressure, Hydraulic Line, 3000 PSI, 5 Micron Absolute, -65° to +275°F,

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER F14 Aircraft, 30 GPM

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